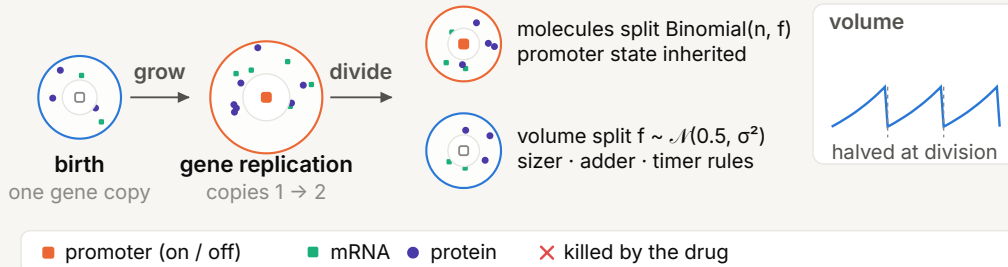


a One cell: stochastic kinetics in a growing, dividing volume

Propensities scale with volume, so concentrations are invariant under growth.



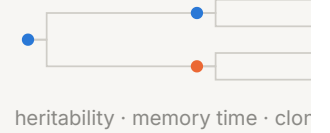
b Population under drug, w

Dose schedules and state-cell at every step, on a free

dose $d(t)$

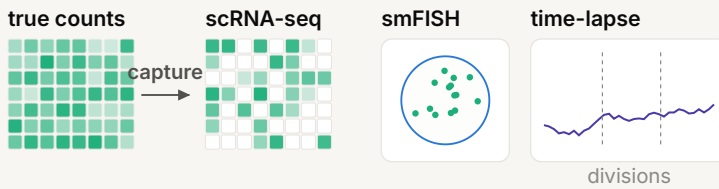
pulsed · piecewise · bolus PK

lineage record: ancestry, st



c From molecules to measurements

One true state, three kinds of measurement.



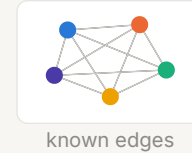
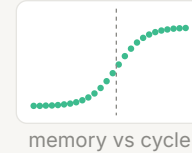
output: CSV · AnnData · Newick · lineage tables

d What the forward model is used for

memory genes

network inference

parameters



ground truth for benchmarks · ABC-SMC · telegraph like

e A complete experiment in twelve lines

```
using Biomodelling, Random
# a resistance gene whose promoter switches more slowly than the cell divides
model = telegraph_model(k_on=0.005, k_off=0.005, k_tx=30.0, k_dm=1.0, k_tl=4.0, k_dp=0.2)
pert = Perturbation(PiecewiseDose([0.0, 100.0], [0.0, 1.0]));
effects = [DeathHazard(h_max=0.5, EC50=0.3, protect=:protein, K=60.0)]
st = PopulationSettings(dt=0.1, growth=ExponentialGrowth(log(2)/20), size_control=Sizer(2.0),
replication=Replication(0.5), control=FreeGrowth())
res = simulate_population(model, x0, 1000, (0.0, 200.0); settings=st, perturbation=pert, rng=Xoshiro(1))

heritability(res, :protein; relation=:sisters) # expression memory across divisions
kill_curve(res); time_to_progression(res) # treatment outcomes
sequence(sample_cells(res; n=500).counts, SeqProtocol(capture=0.15)).Y # synthetic scRNA-seq
```